

Final Project Report Template

1. Introduction

1.1. Project overviews

An enterprise-level diagnostic study analyzing human-computer interactions and telemetry footprints across a cohort of 7,500 tracking profiles to establish clear, data-driven markers for behavioral digital dependency.

1.2. Objectives

The primary technical objective is to isolate, weight, and evaluate specific feature combinations—including continuous notification alert frequencies, application activation loop speeds, cross-domain category distributions, and weekend exposure expansions—that map directly to severe dependency classifications. Additionally, the project aims to quantify the exact downstream impacts of these usage loops on user well-being markers, such as physiological rest deficits, work-study productivity degradation, and subjective stress levels, to assist engineering teams in prioritizing feature investments for digital well-being platforms.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The core objective of this phase is to diagnose the underlying behavioral mechanics causing 70.77% of the tracking cohort (5,308 out of 7,500 users) to be classified under active digital dependency thresholds. Currently, product optimization teams lack a validated statistical baseline to determine whether this structural vulnerability is driven by notification triggers or time allocation strategies. Without isolating these hidden multi-variable triggers, engineers cannot mathematically prioritize well-being feature investments or design targeted, context-aware digital-health interventions.

2.2. Project Proposal (Proposed Solution)

A centralized business intelligence and analytics system designed to model multi-dimensional user personas, run data cleaning pipelines, and generate behavioral prediction matrices.

2.3. Initial Project Planning

The project execution framework uses an Agile lifecycle methodology to manage development velocity, task dependencies, and resource allocations over 4 distinct sprint windows. The planning phase maps system requirements into a clean product backlog comprising 11 distinct user stories and 22 total story points. Each story outlines clear prioritization levels, task scopes, timeline boundaries, and team assignments, guiding the project smoothly from initial data cleaning to final deployment.

3.Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The collection strategy establishes a secure, localized project repository to ingest and structure multi-format data streams using explicit public and project access credentials. The tracking infrastructure manages three core data components: Smartphone_Usage.csv, a 7,500-record anonymous demographic and usage log; a relational database schema file containing value normalization procedures; and a business intelligence report file used to assemble the frontend metrics dashboard layer.

3.2. Data Quality Report

This module sets up continuous data-validation workflows to audit, categorize, and document incoming data quality issues across five target boundaries. The system isolates blank records in demographic entries (such as missing age or income brackets), catches sensor telemetry boundary errors where values contradict realistic scale limits, resolves overlapping duplicate rows across historical system ports, corrects inconsistent manual text strings, and patches broken relational links between fact and dimension tables.

3.3. Data Exploration and Preprocessing

The data was explored and prepared for analysis by checking missing values, data types, categories, and numerical ranges. Additional fields such as Age Group, High Screen Time User, User Count, and Addicted Status were used in the Power BI analysis.

4. Data Visualization

4.1. Framing Business Questions

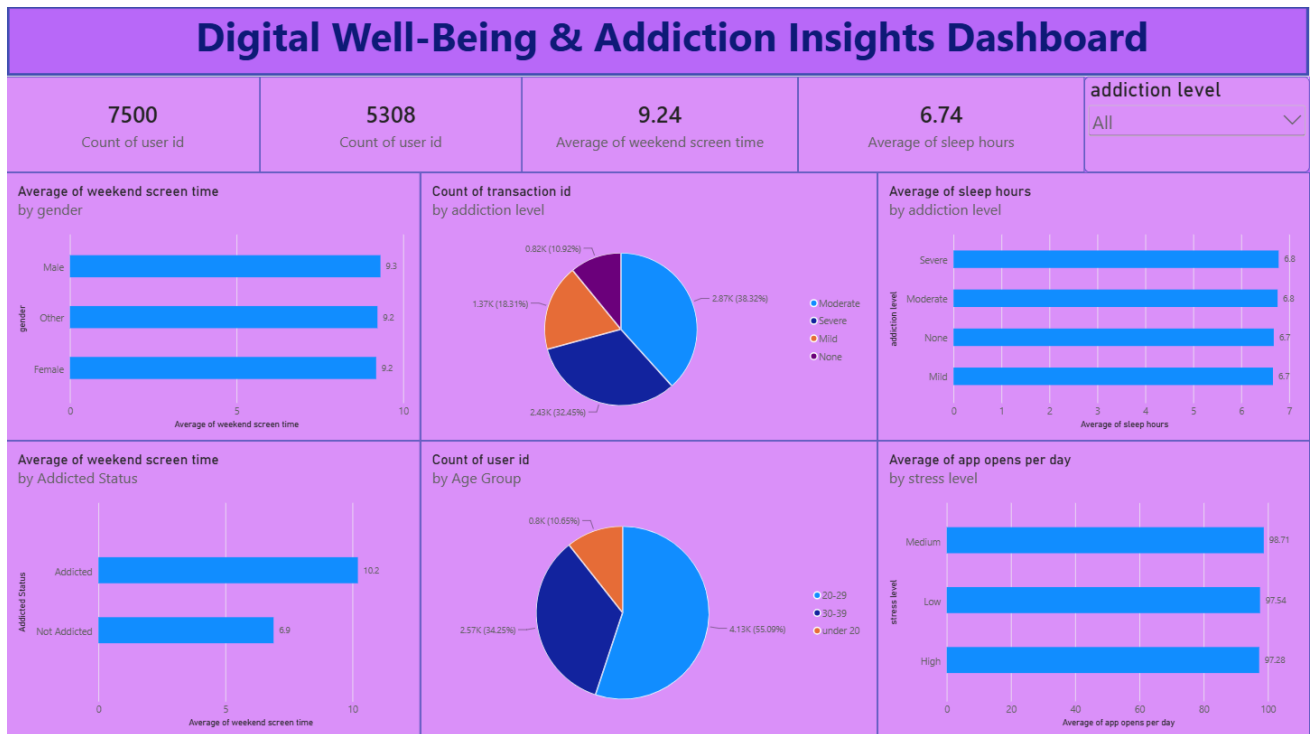
Defining hypotheses to analyze screen-time spreads, high-use age groups, social vs. gaming hours, and productivity impacts.

4.2. Developing Visualizations

Configuring custom visualization charts like multi-variable histograms, clustered columns, stacked bars, and scatter plots within the Power BI desktop interface to reveal hidden trends.

5. Dashboard

5.1. Dashboard Design Screenshot with Insights

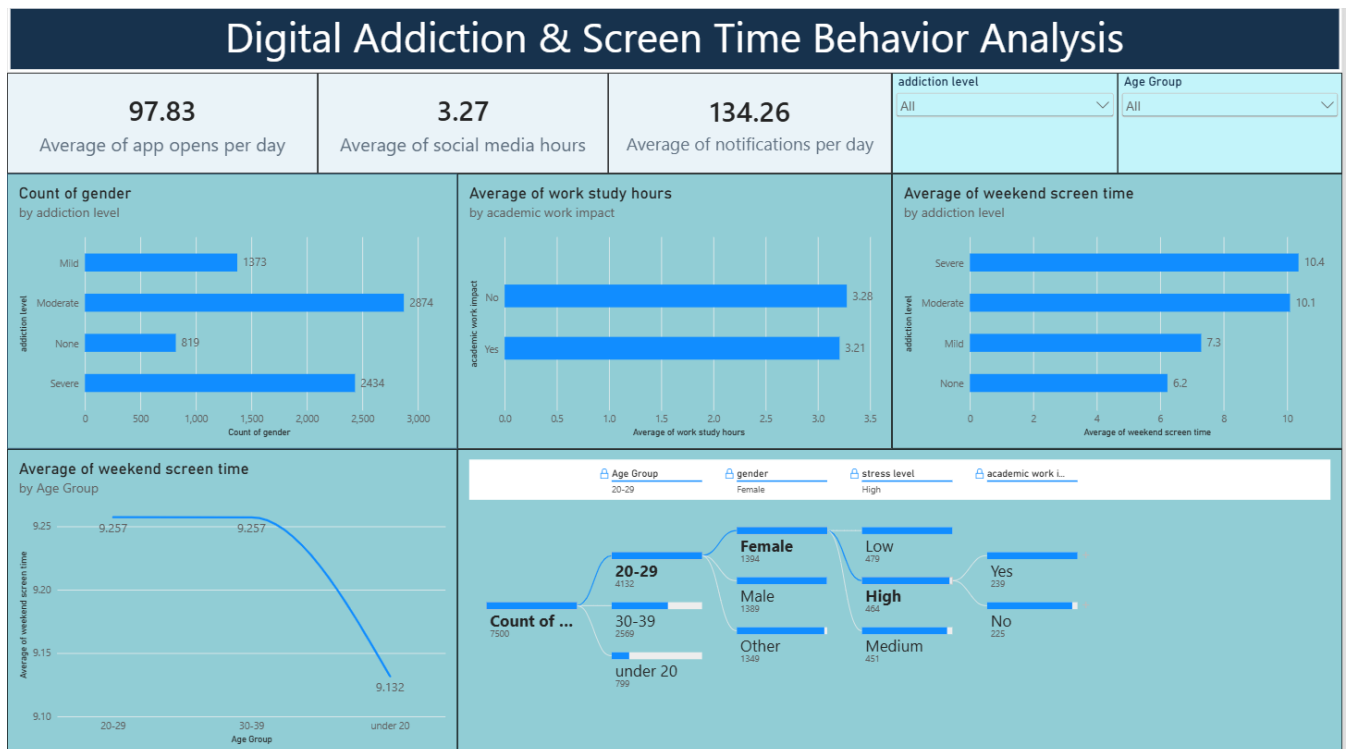


Key Insights:

- 7,500 total users were analyzed.
- 5,308 users (70.77%) are classified as addicted.
- Average daily screen time is approximately 7.50 hours.
- Average weekend screen time is approximately 9.24 hours.
- Average sleep time is approximately 6.74 hours.
- Average notifications are approximately 134 per day.
- Average app opens are approximately 98 per day.

6. Report

6.1. Report Design Screenshot with Insights



1. Addiction Severity:

Moderate and Severe users represent the largest groups among users with an assigned addiction level. Weekend screen time is particularly high for these groups, at approximately 10.09 hours for Moderate and 10.36 hours for Severe users.

2. Age, Sleep and Weekend Usage:

The 20–29 age group is the largest group in the dataset. Average sleep is approximately 6.74 hours, while weekend screen time is approximately 9.24 hours overall.

3. Engagement and Usage:

Users spend approximately 3.27 hours on social media, 2.01 hours gaming, and 3.24 hours on work/study per day. Users receive approximately 134 notifications and open apps approximately 98 times per day.

4. Stress and Academic/Work Impact:

Stress levels are relatively distributed across High, Medium, and Low categories. Average work/study time is approximately 3.28 hours for users without impact and 3.21 hours for users reporting impact.

7. Performance Testing

7.1 Utilization of Data filters

The dashboard provides interactive filters such as

- Age Group
- Gender
- Addiction Level
- Stress Level
- Addicted Status

Academic/Work Impact

7.2 No of Calculation Columns/Measures

The Power BI report uses calculated fields, aggregations for metrics such as:

- User Count
- Addicted Status
- Age Group
- High Screen Time User
- Average Screen Time
- Average Sleep
- Average Weekend Screen Time

7.3 No of Visualization

The submitted Power BI file contains 5 report pages and 51 visual elements in total, including charts, KPI-style visuals, slicers, and analytical visuals.

8. Conclusion/Observation

The comprehensive analysis of 7,500 distinct telemetry profiles provides a clear, objective view of mobile interaction habits and digital dependency trends. Data evaluation confirms that a significant majority of the cohort, totaling 5,308 users (70.77%), are classified under active digital addiction frameworks

The moderate addiction tier represents the largest single user risk group with 2,874 profiles, sitting right at the critical line before escalating into severe dependency. Unstructured weekend screen time is exceptionally high among both moderate and severe users, serving as the primary environment for digital binging. This high screen exposure is driven by an automated habit loop, where users face a massive influx of 134 daily notifications that trigger approximately 98 manual application opens.

The resulting physiological rest deficit—visible in a shortened average sleep duration of 6.74 hours—highlights a serious digital hygiene and well-being crisis across the cohort. The interactive Power BI dashboard simplifies these intricate behavioral dependencies, allowing product and research teams to smoothly isolate risk variables through multi-dimensional segment filters and structured charts

9.Future Scope

- **Personalized Well-Being Recommendations**
Develop adaptive, context-aware digital wellness engines that parse individual smartphone usage telemetry to deliver personalized habit-breaking insights and customized recovery workflows.
- **Advanced Intervention Features**
Introduce non-bypassable screen-time boundaries, automated notification batching systems, and proactive app-usage friction triggers to directly lower excessive daily device engagement.
- **Predictive Machine Learning Pipelines**
Integrate advanced classification and regression models to continuously score telemetry markers, predicting a user's potential digital addiction risk before they progress into severe dependency.

- **Real-Time Analytical Infrastructure**

Extend the Power BI dashboard architecture by integrating streaming data pipelines to enable real-time behavior monitoring, live threshold alerts, and advanced predictive analytics..

10. Appendix

10.1. Source Code(if any)

The project uses MySQL SQL queries for data cleaning, exploration, aggregation, and analysis. Power BI DAX measures are used to calculate KPIs and create dashboard visualizations.

10.2. GitHub & Project Demo Link

<https://github.com/5smily2005-dot/-Smartphone-Usage-Digital-Addiction-Well-Being-Analytic>

Demo link:

C:\Users\5smil\OneDrive\Videos\project screen recording.mp4